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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/269,741	06/26/2023	Haruki Tanaka	S1459.72785US00	1621

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Sony Group Corp / Europe Limited (S1459/S1981)
c/o Wolf, Greenfield & Sacks, P.C.
600 Atlantic Avenue
Boston, MA 02210-2206

EXAMINER	
HSU, AMY R	

ART UNIT	PAPER NUMBER
2638	

NOTIFICATION DATE	DELIVERY MODE
02/27/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

Patents_eOfficeAction@WolfGreenfield.com
Sony_eOfficeAction@WolfGreenfield.com

Office Action Summary	Application No. 18/269,741	Applicant(s) Tanaka, Haruki	
	Examiner AMY R HSU	Art Unit 2638	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 6/26/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-13 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-4 and 9-13 is/are rejected.

8) ☑ Claim(s) 5-8 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 6/26/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 102

2. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

3. **Claims 1-3, 9, 12-13** are rejected under 35 U.S.C. 102(a)(1) as being anticipated by U.S. Patent Application Publication No. 2018/0098033 to Mleczko et al. ("Mleczko").

Regarding **claim 1**, Mleczko teaches a sensor module, comprising:

a sensor element (*reference number 24, image sensor, see Figure 2 and paragraph [0019]*);

a first case that includes an opening end including a first welded region and accommodates therein the sensor element (*reference number 30 is a first case that includes an opening, where the opening faces to the left as seen in Figure 2 and accommodates the sensor 24 within it, see also Figure 3*);

a second case that includes a joining surface including a second welded region, the second welded region being welded to the first welded region to form a welded portion (*reference number 32 is a second case and includes a joining surface which is*

the surface that faces towards reference number 30 in Figure 2; paragraph [0024] teaches that the joint between the front camera housing and the rear camera housing may be closed via laser welding); and

a groove that is formed in at least one of the first welded region or the second welded region (*the groove is seen in Figures 2 and 3 along the inner edges of reference number 32. Figure 3 shows the edge of reference number 30 inserted into the groove of 32)*.

Regarding **claim 2,** Mleczko teaches the sensor module according to claim 1, wherein the groove is formed over an entire circumference of the first welded region or the second welded region (*Figure 2 shows the groove is formed over the entire outer perimeter of the housing 32*). Note that circumference is interpreted to mean around the outline or perimeter and does not require a circular or elliptical shape since Applicant's drawings show the groove to be formed around the outside edges of a square shaped device.

Regarding **claim 3,** Mleczko teaches the sensor module according to claim 2, wherein the groove is formed continuously over the entire circumference of the first welded region or the second welded region (*see the groove along the outer perimeter of housing 32 in Figure 2)*.

Regarding **claim 9,** Mleczko teaches the sensor module according to claim 2, wherein the opening end further includes a stepped surface that is formed outside of the first welded region in parallel to the first welded region, the joining surface further

includes a parallel surface that is formed outside of the second welded region in parallel to the second welded region, and the stepped surface and the parallel surface face each other via a gap (*the cross section of the welded connected parts between reference number 30 and reference number 32 is seen in Figure 3 and shows the stepped surface formed outside the first welded region of 30 and the joining surface outside the second welded region of 32).*

Regarding claim 12, Mleczko teaches the sensor module according to claim 1, wherein the sensor element is an imaging device (*the sensor module is an imaging sensor or camera, see paragraph [0018]).*

Regarding claim 13, Mleczko teaches the sensor module according to claim 1, wherein the first case and the second case are joined to each other using laser welding (*the front housing and rear housing are joined using laser welding, see paragraph [0024]).*

Claim Rejections - 35 USC § 103

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. **Claim 4** is rejected under 35 U.S.C. 103 as being unpatentable over Mleczko as applied to claim 1 above, and further in view of U.S. Patent Application Publication No. 2020/0174343 to Li et al. ("Li").

Regarding **claim 4,** Mleczko teaches the sensor module according to claim 2, but is silent on wherein the groove includes a plurality of first grooves formed in parallel over the entire circumference of the first welded region or the second welded region.

Different variations and elaborations on a groove that connects two parts of a housing are known in the prior art. For example, Li teaches groove extending along the bottom edge of a cover and includes multiple grooves, as seen in Figure 4.

It would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains to modify the teaching of Mleczko with that of Li to substitute a variation of the groove with a plurality of grooves to provide a more stable engagement, as is seen in Li, paragraph [0043].

6. **Claim 10-11** are rejected under 35 U.S.C. 103 as being unpatentable over Mleczko as applied to claim 1 above, and further in view of U.S. Patent Application Publication No. 2014/0037914 to Fujimoto ("Fujimoto").

Regarding claim 10, Mleczko teaches the sensor module according to claim 1, but is silent on wherein the first case is made of a resin material that absorbs laser light of the predetermined wavelength, and the second case is made of a resin material that transmits the laser light of the predetermined wavelength.

Similarly to Mleczko, Fujimoto also teaches using laser welding and further teaches welding a resin part that transmits laser light with a predetermined wavelength

and a resin part that absorbs laser light with the same wavelength and using welding to join the parts with laser light (*see paragraph [0002]).*

It would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains to modify the teaching of Mleczko with that of Fujimoto to use resin materials that allow for laser welding as previously described in the prior art.

Regarding claim 11, Mleczko in view of Fujimoto teach the sensor module according to claim 10, Fujimoto further teaches wherein the groove is filled with the resin material melted from the first case or the second case (*laser light is applied to abutting surfaces and passed through a lid body and is absorbed at the surface of the part to be adjoined. The heat from the laser light causes a joining portion to be melted so that when the laser light stops the joining portion resolidifies to complete welding in the joint portion, see paragraph [0034])*.

It would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains to modify the teaching of Mleczko with that of Fujimoto to use resin materials with properties allow for laser heat melting and welding.

Allowable Subject Matter

7. **Claims 5-8** are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

8. Any inquiry concerning this communication or earlier communications from the examiner should be directed to AMY R HSU whose telephone number is (571)270-3012. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Lin Ye can be reached on (571)272-7372. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

AMY R. HSU
Examiner
Art Unit 2664

/AMY R HSU/
Primary Examiner, Art Unit 2638